New Jersey
Graduation Proficiency
Assessments-Adaptive

Practice Test

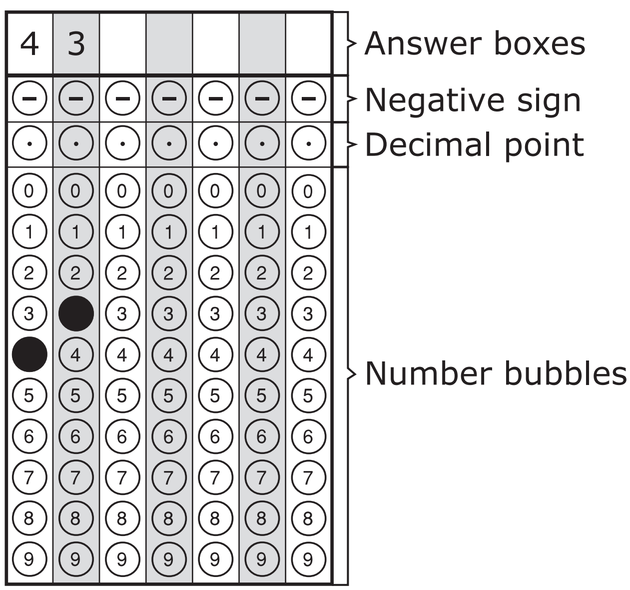
Mathematics

Student Name ______________________________

Directions:

Today, you will take the Mathematics component of the Graduation Proficiency Assessment Practice Test. You will be able to use a calculator. If you do not know the answer to a question, you may go on to the next question. When you finish, you may review your answers and any questions you did not answer.

1. For questions with bubbled responses, fill in the circle next to your answer choice. If you change your answer, make sure you erase your old answer completely. Do not cross out or make any marks on the other choices.
2. For questions with response boxes, write your answer neatly, clearly, and in the space provided.
3. For questions with response grids, write your answer in the answer boxes at the top of the grid.
   - Write only one digit or symbol in each answer box.
   - Use only the options listed below each box.
   - Be sure to write a negative sign or decimal point in the answer box if it is part of the answer.



GO ON ▶

1. The number of visitors, in millions, to Bryce Canyon National Park from 2010 to 2014 can be modeled by the expression $1.281(1.024)^t$, where t is the number of years since 2010.

What is the factor, after t years, by which the number of visitors has grown since 2010?

Ⓐ 1.281

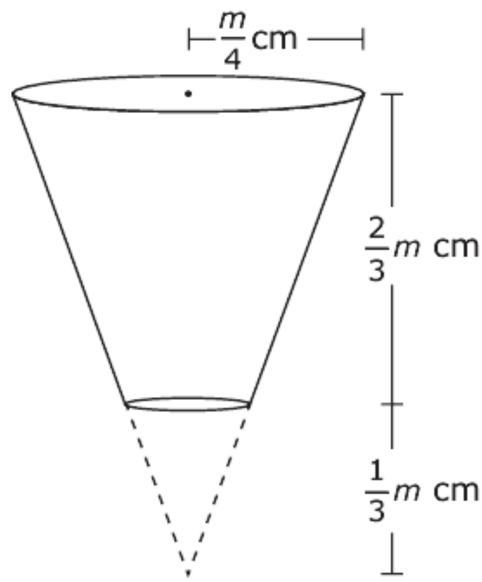
Ⓑ 1.024

Ⓒ $(1.024)^t$

Ⓓ $1.281(1.024)^t$

GO ON ▶

2. A popcorn container is made from a cone with a portion of the cone removed and sealed, as shown. The removed portion of the cone has a height of $\frac{1}{3}m$.



Create an expression, using m, that represents the volume, in cubic centimeters, of the popcorn container.

3. This item has two parts.

Some friends spent a total of $12.00 on popcorn and drinks at the movie theater. A bucket of popcorn cost $2.00 and a drink cost $1.50.

Part A.

Create an equation to represent the relationship between the number of buckets of popcorn, x, and the number of drinks, y, the friends bought for $12.00.

Part B.

The friends bought 4 drinks.

How many buckets of popcorn did they buy?

4. A quadratic equation is shown.

$$0 = x^2 - 3x - 4$$

Which value is a solution to this equation?

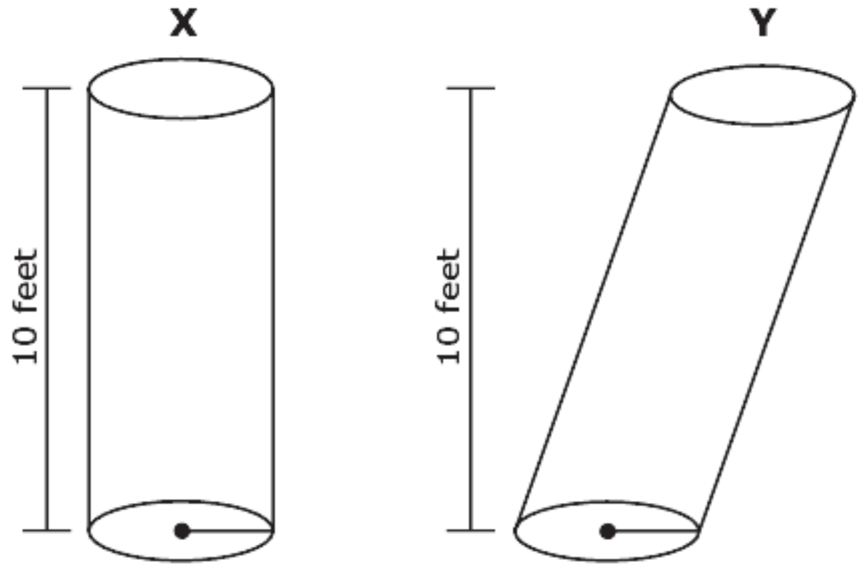
Ⓐ 1

Ⓑ 2

Ⓒ 3

Ⓓ 4

5. Two cylinders, X and Y, are shown. Each cylinder has a height of 10 feet.



Which statement about these cylinders is true?

Ⓐ The volumes of the two cylinders are always equal because they have the same height.

Ⓑ The volume of cylinder Y is always greater because the slant height of cylinder Y is greater than the height of cylinder X.

Ⓒ The relationship between the volumes of the two cylinders cannot be determined because the slant height of cylinder Y is not given.

Ⓓ The relationship between the volumes of the two cylinders cannot be determined because the radii of the two cylinders are not given.

6. A function is shown.

$$h(t) = -t^2 + 10t - 16$$

For which interval of t-values is the function both positive and increasing?

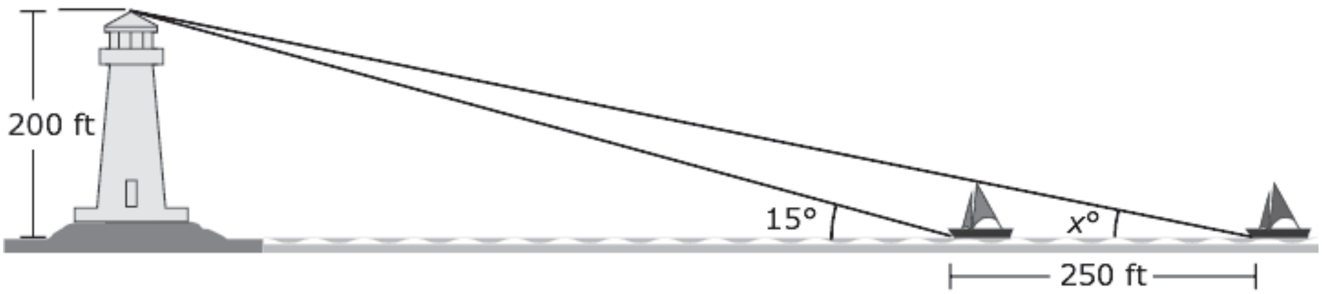
Ⓐ $t < 5$

Ⓑ $t > 8$

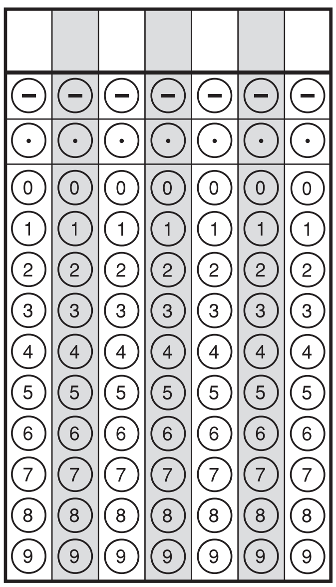
Ⓒ $2 < t < 5$

Ⓓ $5 < t < 8$

7. Two boats are traveling toward a lighthouse that is 200 feet (ft) above sea level at its top. When the two boats and the lighthouse are collinear, the boats are exactly 250 feet apart and the boat closest to the lighthouse has an angle of elevation to the top of the lighthouse of 15°, as shown.



What is the value of x, rounded to the nearest hundredth?

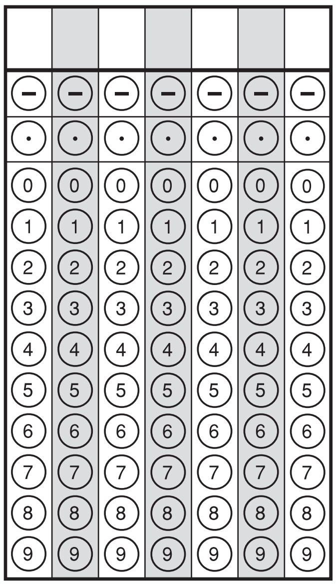


GO ON ▶

8. Hannah has a cone made of steel and a cone made of granite.

- Each cone has a height of 10 centimeters and a radius of 4 centimeters.
- The density of steel is approximately 7.75 grams per cubic centimeter.
- The density of granite is approximately 2.75 grams per cubic centimeter.

What is the difference, to the nearest gram, of the masses of the cones?



9. In 2015, Macon County had a population of 53,792. The population increases by 2.5% annually.

Which function can be used to model the population t years after 2015?

Ⓐ $f(t) = 1.025t + 53{,}792$

Ⓑ $f(t) = 1.25t + 53{,}792$

Ⓒ $f(t) = 53{,}792(1.025)^t$

Ⓓ $f(t) = 53{,}792(1.25)^t$

10. Steven constructs an equilateral triangle inscribed in circle *P*. His first three steps are shown.

1. He creates radius $\overline{PQ}$ using a point *Q* on the circle.
2. Using point *Q* as the center and the length of $\overline{PQ}$ as a radius, he uses a compass to construct an arc that intersects the circle at *R*.
3. Using point *R* as the center and the length of $\overline{PQ}$ as a radius, he uses a compass to construct an arc that intersects the circle at *S*.

What should be Steven's next step in constructing the equilateral triangle?

Ⓐ Draw line segments connecting the points *Q*, *R*, and *S* to construct ΔQRS.

Ⓑ Draw line segments connecting the points *P*, *R*, and *S* to construct ΔPRS.

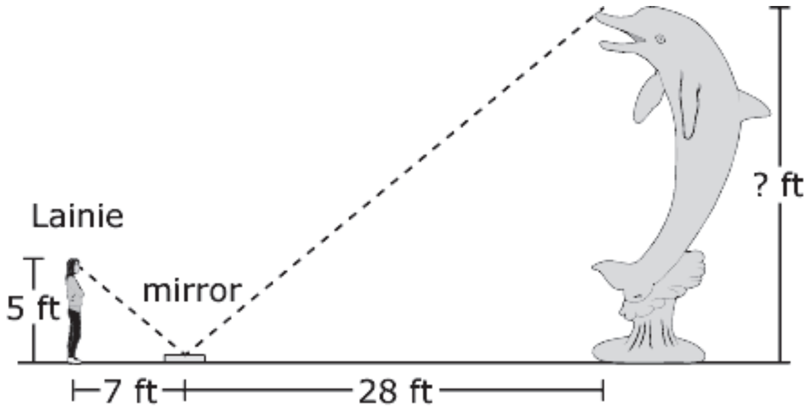
Ⓒ Construct an arc intersecting the circle by using point *S* as the center and the length of $\overline{PQ}$ as a radius.

Ⓓ Construct an arc intersecting the circle by using point *P* as the center and the length of $\overline{PQ}$ as a radius.

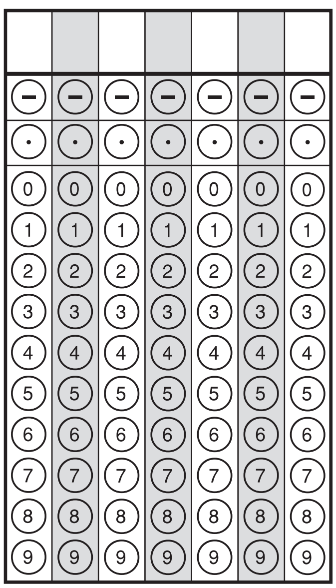
11. A rectangular park has an area of 250 square feet. The length of the park is 7 feet more than twice the width, *w*, of the park.

Create an equation in terms of *w* to model this situation.

12. Lainie wants to calculate the height of a sculpture. She places a mirror on the ground so that when she looks into the mirror she sees the top of the sculpture, as shown.



What is the height, in feet, of the sculpture?



13. A function is shown, where b is a real number.

$$f(x) = x^2 + bx + 144$$

The minimum value of the function is 80.

Create an equation for an equivalent function in the form $f(x) = (x - h)^2 + k$.

$f(x) =$ $\boxed{}$

14. The model $n(t) = 2^t$ represents the number of bacteria in a petri dish after t hours, where $t = 0$ represents the time when the bacteria were first put into the dish.

What is the correct value and interpretation of $n(8)$?

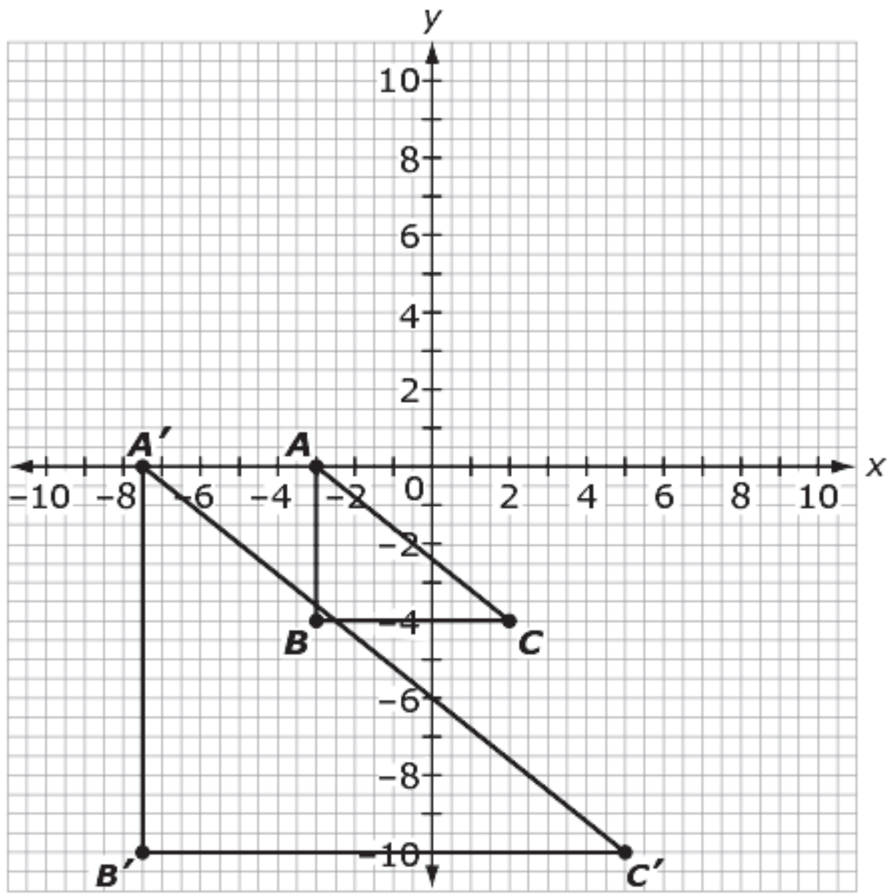
Ⓐ $n(8) = 256$, so after 8 hours there are 256 bacteria.

Ⓑ $n(8) = 256$, so after 256 hours there are 8 bacteria.

Ⓒ $n(8) = 3$, so after 8 hours there are 3 bacteria.

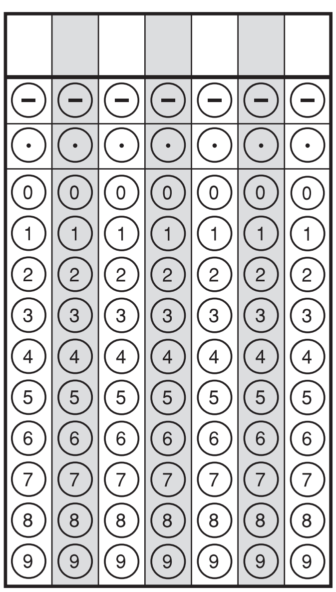
Ⓓ $n(8) = 3$, so after 3 hours there are 8 bacteria.

15. Triangle ABC is dilated with a scale factor of k and a center of dilation at the origin to obtain triangle $A'B'C'$.



What is the scale factor?

k =



GO ON ▶

16. The gravitational potential energy of an object is given by the formula $P = mgh$.

Which equation is correctly solved for the height, h?

Ⓐ $h = P + mg$

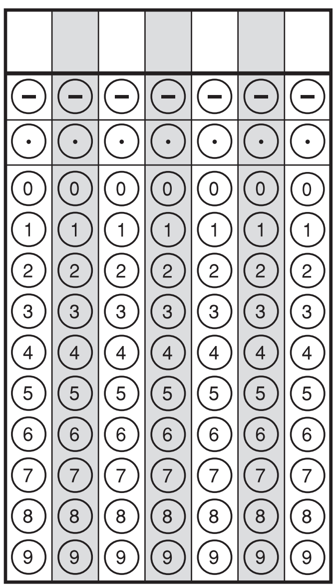
Ⓑ $h = P - mg$

Ⓒ $h = \frac{P}{mg}$

Ⓓ $h = Pmg$

17. The function $f(t) = -16t^2 + 20t + 4$ gives the height of a ball, in feet, t seconds after it is tossed.

What is the average rate of change, in feet per second, over the interval [0.75, 1.25]?



GO ON ▶

18. What is the exact perimeter of a parallelogram with vertices at (3, 2), (4, 4) and (6, 1)?

19. The function $h(p)$ represents the cost for a company to manufacture p posters.

What is the domain of the function?

- (A) all integers
- (B) all real numbers
- (C) all non-negative integers
- (D) all positive rational numbers

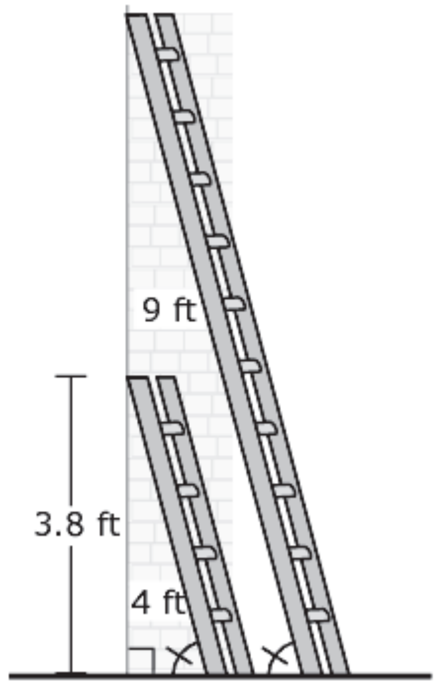
20. A square is rotated about its center.

Select all of the angles of rotation that will map the square onto itself.

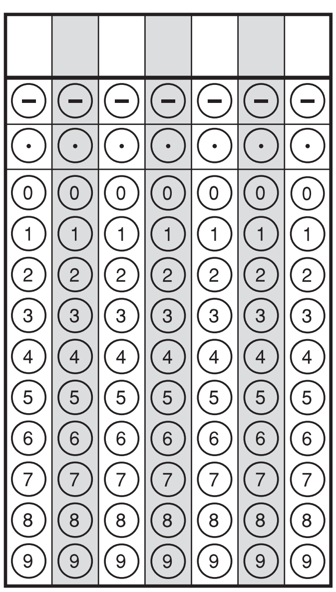
- (A) 45 degrees
- (B) 60 degrees
- (C) 90 degrees
- (D) 120 degrees
- (E) 180 degrees
- (F) 270 degrees

GO ON ▶

21. A 9-foot (ft) ladder and a 4-foot ladder are leaning against a house. The two ladders create angles of the same measure with the ground. The 4-foot ladder has a height of 3.8 feet against the house.



What is the height, in feet, of the 9-foot ladder against the house?



22. Select all of the values of a correlation coefficient that suggest a strong linear relationship between two variables.

Ⓐ 0.8

Ⓑ 0.4

Ⓒ 0

Ⓓ –0.1

Ⓔ –0.9

23. A system of equations is shown.

$$4c + 2d = 11$$

$$\frac{7}{2}d = 41 - 22c$$

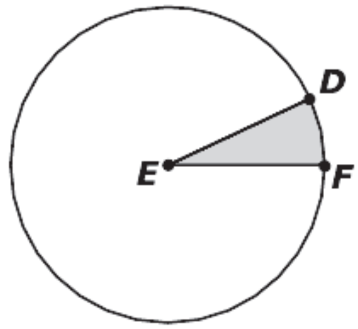
What is the solution to the system?

c =

d =

GO ON ▶

24. Circle E is shown, where $\angle E$ measures 0.44 radians and EF = 7 inches.



What is the area of the shaded region, to the nearest hundredth of a square inch?

25. All corresponding sides and angles of ΔRST and ΔDEF are congruent.

Select all of the statements that must be true.

- Ⓐ There is a reflection that maps $\overline{RS}$ to $\overline{DE}$.
- Ⓑ There is a dilation that maps ΔRST to ΔDEF.
- Ⓒ There is a translation followed by a rotation that maps $\overline{RT}$ to $\overline{DF}$.
- Ⓓ There is a sequence of transformations that maps ΔRST to ΔDEF.
- Ⓔ There is not necessarily a sequence of rigid motions that maps ΔRST to ΔDEF.

GO ON ▶

26. Joshua draws two squares. The area of square M is $\frac{3}{4}$ the area of square N.

Create an equation that relates the side length of square M, *s*, and the area of square N, *A*.

27. Create the equation of a line that is perpendicular to $2y = 14 + \frac{2}{3}x$ and passes through the point $(-2, 8)$.

GO ON ▶

28. Juan records the number of hot dogs sold and the number of cars that drive by a hot dog stand each hour. His data are shown in the table.

Juan's Data

Hour	Number of Hot Dogs Sold	Number of Cars
1	30	73
2	9	32
3	21	56
4	0	11
5	24	62

Based on his data, which conclusion can Juan make?

Ⓐ An increase in cars is associated with a decrease in hot dog sales.

Ⓑ An increase in cars is associated with an increase in hot dog sales.

Ⓒ An increase in cars causes a decrease in hot dog sales.

Ⓓ An increase in cars causes an increase in hot dog sales.

STOP

Paper Practice Test Answer and Alignment Document: Math

NJGPA Mathematics

Item Sequence	Answer Key	Standards Alignment
1	C	A.SSE.A.1a
2	$\frac{1}{3}m\left(\frac{m}{4}\right)^2\pi - \frac{1}{3}\left(\frac{m}{12}\right)^2\left(\frac{m}{3}\right)\pi$, or any equivalent expression	G.MG.A.1
3	Part A: $2x + 1.5y = 12$, or any equivalent equation Part B: 3, or any equivalent value	A.CED.A.2
4	D	A.REI.B.4b
5	D	G.GMD.A.1
6	C	F.IF.B.4
7	any value between 11.349 and 11.355, inclusive	G.SRT.C.8
8	any value between 837 and 838.1, inclusive	G.MG.A.2
9	C	F.LE.A.2
10	C	G.CO.D.13
11	$2w^2 + 7w = 250$, or any equivalent equation	A.CED.A.1
12	20, or any equivalent value	G.SRT.B.5
13	$(x-8)^2 + 80$ or $(x+8)^2 + 80$, or any equivalent expressions	F.IF.C.8a
14	A	F.IF.A.2
15	2.5, or any equivalent value	G.SRT.A.1b
16	C	A.CED.A.4
17	–12, or any equivalent values	F.IF.B.6
18	$2\sqrt{10} + 2\sqrt{5}$, or $2\sqrt{13} + 2\sqrt{5}$, or $2\sqrt{10} + 2\sqrt{13}$, or any equivalent expression	G.GPE.B.7
19	C	F.IF.B.5
20	C, E, F	G.CO.A.3
21	8.55, or any equivalent value	G.SRT.B.5
22	A, E	S.ID.C.8
23	1.45 and 2.6, or any equivalent values	A.REI.C.6
24	10.78, or any equivalent value	G.C.B.5
25	C, D	G.CO.B.7
26	$s^2 = \frac{3}{4}A$, or any equivalent equation	A.CED.A.2
27	$y = -3x + 2$, or any equivalent equation	G.GPE.B.5
28	B	S.ID.C.9